Homework 6

Problem 1. Order the index (from 1 to 9) of the following functions according to their growth rate, and express this ordering using the asymptotic notation O:

(1) $n \ln n$

(2) $(\ln \ln n)^{\ln n}$

(3) $(\ln n)^{\ln \ln n}$

(4) $n \cdot e^{\sqrt{\ln n}}$

(5) $(\ln n)^{\ln n}$

(6) $n \cdot 2^{\ln \ln n}$

(7) $n^{1+\frac{1}{\ln \ln n}}$

(8) $n^{1+\frac{1}{\ln n}}$

(9) n^2

Solution. Define $\preceq$ as '$i \preceq j$ iff $f_i = O(f_j)$', and we use $i \equiv j$ iff $f_i = \Theta(f_j))$. Then $3 \preceq 8 \preceq 6 \preceq 1 \preceq 4 \preceq 7 \preceq 9 \preceq 2 \preceq 5$.

□

Problem 2.

a) Show that the product of all primes p with $m < p \le 2m$ is at most $\binom{2m}{m}$.

b) (*: this question is optional.)
Using a), prove the estimate $\pi(x) = O(\frac{x}{\ln x})$, where $\pi(x)$ denote the number of primes not exceeding the number x.

Solution.

1. $B = \binom{2m}{m} = \frac{(m+1)\times(m+2)\times\cdots\times(2m)}{1\times2\times\cdots\times m}$. It is easy to find that if p is a prime number and $p \in (m, 2m]$ then $p|B$. Thus $\prod\limits_{m<p\le 2m} p \mid B$. It follows that the upper bound of the products of prime numbers between m and $2m$ is B.

2. There are several ways to prove the second problem.

First proof: Combing a), w.l.o.g. assume n is even and $n = 2m$. It is obvious that

$$B \leq \sum_{i=0}^{2m} \binom{2m}{i} = 4^m$$

With a) we have $\prod_{m<p\leq 2m} p \leq B \leq 4^m$ (p is prime, as above). It follows that

$$\sum_{m<p\leq 2m} \log p \leq m \log 4 = 2m \qquad (\star)$$

Then count the number of primes between m and $2m$, i.e. the number of $p \in (m, 2m]$,

$$\pi(2m) - \pi(m) = \sum_{m<p\leq 2m} 1 \leq \sum_{m<p\leq 2m} \frac{\log p}{\log m} = \frac{1}{\log m}\left(\sum_{m<p\leq 2m} \log p\right) \overset{(\star)}{\leq} \frac{2m}{\log m}.$$

For any given x, there exists $k \geq 1$ such that $x \in (2^{k-1}, 2^k]$.
Finally with the above analysis

$$\pi(x) \leq \pi(2^k) = \sum_{i=1}^{k} \left(\pi(2^i) - \pi(2^{i-1})\right) = O\left(\sum_{j=1}^{k} \frac{2^j}{j}\right) = O\left(\frac{2^k}{k}\right) = O\left(\frac{x}{\ln x}\right).$$

Second proof: Proof by contradiction

□

Problem 3. Two simple graphs $G = (V, E)$ and $G' = (V', E')$. A map $f : V \to V'$. Now if f satisfies:

i) It is a bijective function;

ii) $\{x, y\} \in E$ if and only if $\{f(x), f(y)\} \in E'$;

Then we say that graph G and G' are isomorphic to each other. We use $G \cong G'$ to stand for the isomorphism relation.

Consider the following questions:

1. $G = K_n$ (Recall: K_n is a clique with n vertices), $g : V \to V'$ is a function which only satisfies requirement ii). Prove that G' must contain a subgraph which is a clique with n-vertices.

2. $G = K_{n,m}$ (Recall: $K_{n,m}$ is the so-called complete bipartite graphs), g is the same as in question 1. What will be the simplest G' that is related to G under the new relation.

Solution.

1. Two different vertices in G must be mapped to different vertices in G'. For if $u, v \in V$ are mapped to the same vertex $\omega \in V'$, the edge $\{u, v\} \in E$ cannot be reflected in E', for G' is a simple graph.

2. G' can be just one edge with two incident vertices.

□

Problem 4. How many graphs on the vertex set $\{1, 2, \ldots, 2n\}$ are isomorphic to the graph consisting of n vertex-disjoint edges (i.e. with edge set {{1,2},{3,4},..., {2n-1,2n}}?

Solution. $\frac{(2n\cdot(2n-1))((2n-2)\cdot(2n-3))\cdots(2\cdot 1)}{2^n \cdot n!} = (2n-1)(2n-3)\cdot\cdots\cdot 5\cdot 3.$

□

Problem 5. Construct an example of a sequence of length n in which each term is some of the numbers $1, 2, \ldots, n-1$ and which has an even number of odd terms, and yet the sequence is not a graph score. Show why it is not a graph score.

Solution. E.g. $(1, 1, 3, 3, 4)$. Use the Score theorem to prove that it cannot be a graph score. □